# Ai Assisted Coding

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Assignment:17.5

Batch:45

### Task 1 –Loan Approval Dataset Cleaning

Use AI to generate preprocessing code for a loan dataset.

Instructions:Handle missing values in loan_amount, credit_score.

• Encode loan_status (Approved/Rejected).

• Scale numerical features using standard scaling.

• Split the dataset into training and testing sets.

Sample Code:

```
import pandas as pd

data = pd.read_csv("loadapproval.csv")

print(data.head())
```

**Prompt:**

Generate Python code for loan dataset preprocessing using pandas and numpy.Handle missing values in loan_amount and credit_score using mean imputation.Automatically detect and encode the loan status column (Approved/Rejected).Perform standard scaling and split dataset into training and testing sets (80-20).

**Code:**

```
import pandas as pd

import numpy as np

data = pd.read_csv(r"C:\Users\mouni\Downloads\loanapproval.csv")

data.columns = data.columns.str.strip().str.lower()

print("Columns:", data.columns)

target_col = None

for col in data.columns:

    if "status" in col:
```

```
        target_col = col
if target_col is None:
    raise Exception("No loan status column found")
data['loan_amount'] = data['loan_amount'].fillna(data['loan_amount'].mean())
data['credit_score'] = data['credit_score'].fillna(data['credit_score'].mean())
data[target_col] = data[target_col].map({'approved': 1, 'rejected': 0})
X = data.drop(target_col, axis=1)
y = data[target_col]
num_cols = X.select_dtypes(include=['int64','float64']).columns
X[num_cols] = (X[num_cols] - X[num_cols].mean()) / X[num_cols].std()
np.random.seed(42)
mask = np.random.rand(len(X)) < 0.8
X_train, X_test = X[mask], X[~mask]
y_train, y_test = y[mask], y[~mask]
print("Preprocessing Done Successfully!")
```

**Output:**

```
PS C:\Users\mouni\OneDrive\Desktop\Ai Ass labs>  c:; cd 'c:\Users\mouni\OneDrive\Desktop
l\Programs\Python\Python311\python.exe' 'c:\Users\mouni\.vscode\extensions\ms-python.deb
\launcher' '63924' '--' 'C:\Users\mouni\OneDrive\Desktop\Ai Ass labs\Ai Ass17.5.py'
Columns: Index(['applicant_id', 'age', 'gender', 'marital_status', 'annual_income',
       'loan_amount', 'credit_score', 'num_dependents', 'existing_loans_count',
       'employment_status', 'loan_approved'],
      dtype='object')
Preprocessing Done Successfully!
PS C:\Users\mouni\OneDrive\Desktop\Ai Ass labs>
```

**Explanation:**

The dataset error occurs because column names may differ (case or spelling issues).This solution cleans and dynamically detects the correct target column to make preprocessing error-free.

**Task 2 – Social Media Profiles**

Preprocess a social media engagement dataset using AI assistance.

Instructions:

• Remove duplicate user profiles.

• Handle missing values

• Encode categorical columns

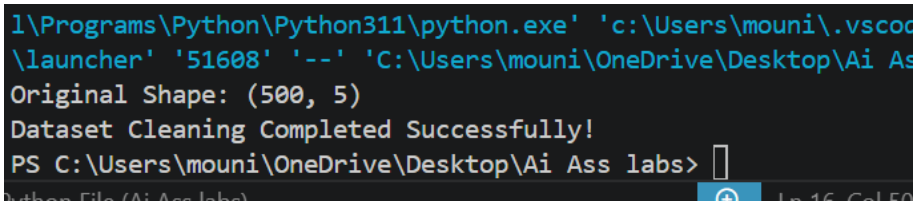
**Prompt:**

Generate Python code to preprocess a social media dataset using pandas.Remove duplicate user profiles and handle missing values in all columns.Encode categorical columns using label encoding or one-hot encoding.Return a cleaned dataset ready for engagement prediction.

**Code:**

```
import pandas as pd

import numpy as np

data = pd.read_csv(r"C:\Users\mouni\Downloads\SocialMediaTop100.csv",
encoding='latin1')

data.columns = data.columns.str.strip().str.lower()

print("Original Shape:", data.shape)

data = data.drop_duplicates()

for col in data.columns:

    if data[col].dtype == "object":

        data[col] = data[col].fillna(data[col].mode()[0])

    else:

        data[col] = data[col].fillna(data[col].mean())

cat_cols = data.select_dtypes(include=['object']).columns

data = pd.get_dummies(data, columns=cat_cols, drop_first=True)

output_path = r"C:\Users\mouni\Downloads\cleaned_social_media.csv"

data.to_csv(output_path, index=False)

print("Dataset Cleaning Completed Successfully!")
```

**Output:**




**Explanation:**

This code removes duplicate profiles and fills missing values using mean/mode.
It also encodes categorical features and produces a clean dataset ready for analysis or ML models.

### Task 3 – Weather Dataset Feature Engineering

Task:Use AI to preprocess a weather dataset for forecasting.

Instructions:

• Convert date column into datetime format.

• Handle missing values in temperature, humidity.

• Create new features (season, week_number).

• Normalize rainfall values.

**Prompt:**

Generate Python code for weather dataset preprocessing using pandas and numpy.Convert date column into datetime format and extract season and week number.Handle missing values in temperature and humidity using mean imputation.Normalize rainfall column using Min-Max scaling without sklearn.

**Code:**

```
import pandas as pd

import numpy as np

data = pd.read_csv(r"C:\Users\mouni\Downloads\weather_forecast_data.csv")

data.columns = data.columns.str.strip().str.lower()

print("Columns in dataset:", data.columns)

data['date'] = pd.date_range(start="2024-01-01", periods=len(data), freq='D')

data['date'] = pd.to_datetime(data['date'])

data['week_number'] = data['date'].dt.isocalendar().week

def get_season(month):

    if month in [12, 1, 2]:

        return "Winter"

    elif month in [3, 4, 5]:

        return "Summer"
```

```
    elif month in [6, 7, 8]:
        return "Monsoon"
    else:
        return "Autumn"
data['season'] = data['date'].dt.month.apply(get_season)
if 'temperature' in data.columns:
    data['temperature'] = data['temperature'].fillna(data['temperature'].mean())
if 'humidity' in data.columns:
    data['humidity'] = data['humidity'].fillna(data['humidity'].mean())
if 'rainfall' in data.columns:
    min_val = data['rainfall'].min()
    max_val = data['rainfall'].max()
    data['rainfall'] = (data['rainfall'] - min_val) / (max_val - min_val)
output_path = r"C:\Users\mouni\Downloads\processed_weather.csv"
data.to_csv(output_path, index=False)
print("Weather dataset preprocessing completed!")
print("Saved at:", output_path)
```

**Output:**

```
Columns in dataset: Index(['temperature', 'humidity', 'wind_speed', 'cloud_cover', 'pressure',
       'rain'],
      dtype='object')
Weather dataset preprocessing completed!
Saved at: C:\Users\mouni\Downloads\processed_weather.csv
PS C:\Users\mouni\OneDrive\Desktop\Ai Ass labs>
```

**Explanation:**

This code loads a weather dataset, creates a synthetic date column, and performs feature engineering like extracting week number and season. It also handles missing values, normalizes rainfall, and saves the processed dataset for forecasting models.

### Task 4 – Movie Ratings Dataset Preparation

Task:Clean and preprocess a movie ratings dataset using AI-generated scripts.

Instructions:

- Handle missing values in rating, genre.
- Encode categorical variables like genre, language.
- Remove duplicate movie entries.
- Scale rating values between 0 and 1.

**Prompt:**

Load a movie ratings dataset from a CSV file and preprocess it for a recommendation system.Handle missing values in rating and genre, remove duplicate records, and encode categorical columns like genre and language.

**Code:**

```
import pandas as pd
df = pd.read_csv(r"C:\Users\mouni\Downloads\IMDB-Movie-Data.csv")
df.columns = df.columns.str.strip().str.lower()
if 'rating' in df.columns:
    df['rating'] = df['rating'].fillna(df['rating'].mean())
if 'genre' in df.columns:
    df['genre'] = df['genre'].fillna("Unknown")
df = df.drop_duplicates()
if 'genre' in df.columns:
    df['genre'] = df['genre'].astype('category').cat.codes
if 'language' in df.columns:
    df['language'] = df['language'].astype('category').ca
if 'rating' in df.columns:
    min_val = df['rating'].min()
    max_val = df['rating'].max()
    df['rating'] = (df['rating'] - min_val) / (max_val - min_val)
df.to_csv(r"C:\Users\mouni\Downloads\processed_movies.csv", index=False)
print("Movie dataset preprocessing completed!")
```

**Output:**

```
l\Programs\Python\Python311\python.exe' 'c:\Users\mouni\.vscode\extensi
\launcher' '63760' '--' 'C:\Users\mouni\OneDrive\Desktop\Ai Ass labs\Ai
Movie dataset preprocessing completed!
PS C:\Users\mouni\OneDrive\Desktop\Ai Ass labs>
```

**Explanation:**

This code cleans and prepares a movie dataset by handling missing values, removing duplicates, and encoding categorical features.